

Lab Assignment 7.4

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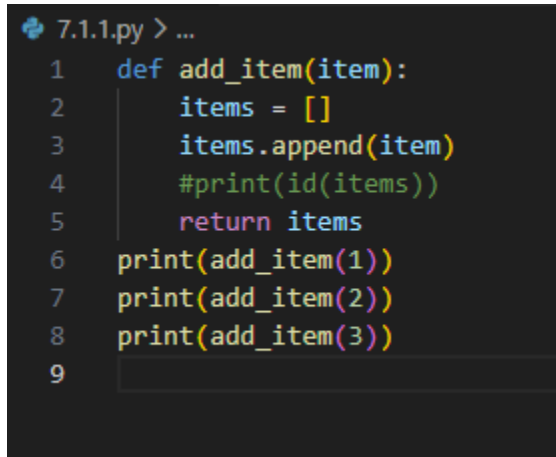
1. (Mutable Default Argument – Function Bug)

Task:

Analyze given code where a mutable default argument causes unexpected behavior. Use AI to fix it.

```
# Bug: Mutable default argument
def add_item(item, items= []):
    items.append(item)
    return items
print(add_item(1))
print(add_item(2))
```

Expected Output: Corrected function avoids shared list bug.



```
7.1.1.py > ...
1  def add_item(item):
2      items = []
3      items.append(item)
4      #print(id(items))
5      return items
6  print(add_item(1))
7  print(add_item(2))
8  print(add_item(3))
9
```

2. (Floating-Point Precision Error)

Task:

Analyze given code where floating-point comparison fails. Use AI to correct with tolerance.

```
# Bug: Floating point precision issue
def check_sum():
    return (0.1 + 0.2) == 0.3
print(check_sum())
```

Expected Output: Corrected function

7.4.2.py > check_sum

```
1
2 def check_sum():
3     return abs(0.1 + 0.2 - 0.3) < 1e-15
4 print(check_sum())
5
6
7 def check_sum():
8     return round(0.1 + 0.2, 1) == 0.3
9 print(check_sum())
10
11 import math
12 def check_sum():
13     return math.isclose(0.1 + 0.2, 0.3, rel_tol=1e-9)
14 print(check_sum())
```

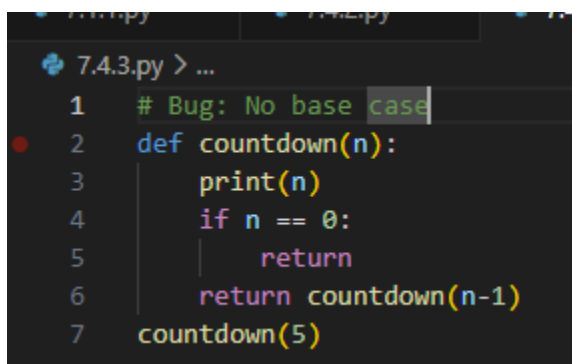
3. (Recursion Error – Missing Base Case)

Task:

Analyze given code where recursion runs infinitely due to missing base case. Use AI to fix it.

```
# Bug: No base case
def countdown(n):
    print(n)
    return countdown(n-1)
countdown(5)
```

Expected Output: Correct recursion with stopping condition.



```
7.4.3.py > ...
1 # Bug: No base case
2 def countdown(n):
3     print(n)
4     if n == 0:
5         return
6     return countdown(n-1)
7 countdown(5)
```

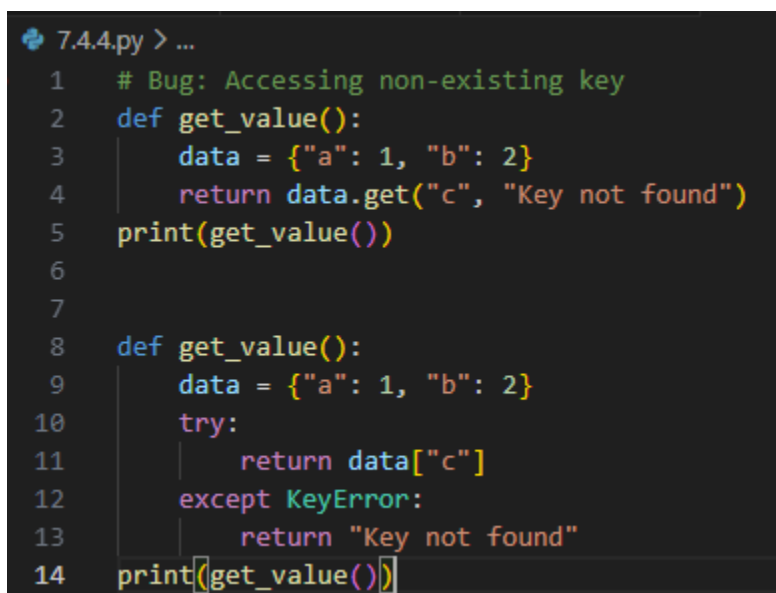
4. (Dictionary Key Error)

Task:

Analyze given code where a missing dictionary key causes error. Use AI to fix it.

```
# Bug: Accessing non-existing key
def get_value():
    data = {"a": 1, "b": 2}
    return data["c"]
print(get_value())
```

Expected Output: Corrected with .get() or error handling.



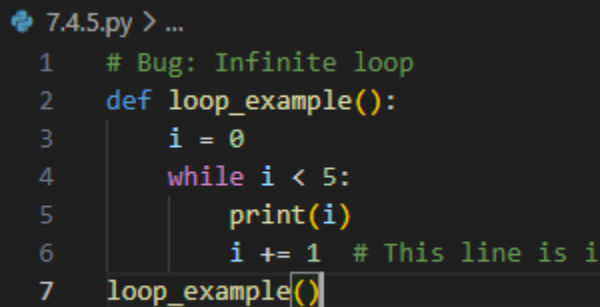
```
7.4.4.py > ...
1 # Bug: Accessing non-existing key
2 def get_value():
3     data = {"a": 1, "b": 2}
4     return data.get("c", "Key not found")
5 print(get_value())
6
7
8 def get_value():
9     data = {"a": 1, "b": 2}
10    try:
11        return data["c"]
12    except KeyError:
13        return "Key not found"
14 print(get_value())
```

5. (Infinite Loop – Wrong Condition)

Task: Analyze given code where loop never ends. Use AI to detect and fix it.

```
# Bug: Infinite loop
def loop_example():
    i = 0
    while i < 5:
        print(i)
```

Expected Output: Corrected loop increments i.



```
7.4.5.py > ...
1 # Bug: Infinite loop
2 def loop_example():
3     i = 0
4     while i < 5:
5         print(i)
6         i += 1 # This line is i
7 loop_example()
```

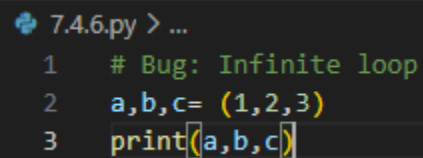
6. (Unpacking Error – Wrong Variables)

Task:

Analyze given code where tuple unpacking fails. Use AI to fix it.

```
# Bug: Wrong unpacking
a, b = (1, 2, 3)
```

Expected Output: Correct unpacking or using _ for extra values.



```
7.4.6.py > ...
1 # Bug: Infinite loop
2 a,b,c= (1,2,3)
3 print(a,b,c)
```

7. (Mixed Indentation – Tabs vs Spaces)

Task: Analyze given code where mixed indentation breaks execution. Use AI to fix it.

```
# Bug: Mixed indentation
def func():
    x = 5
    y = 10
    return x+y
```

Expected Output : Consistent indentation applied.

```

7.4.7.py > ...
1  # Bug: Mixed indentation
2  def fun():
3      x = 5
4      y = 10
5      return x + y
6  print(fun())

```

8. (Import Error – Wrong Module Usage)

Task:

Analyze given code with incorrect import. Use AI to fix.

```

# Bug: Wrong import
import maths
print(maths.sqrt(16))

```

Expected Output: Corrected to import math

```

7.4.8.py
1  # Bug: Wrong import
2  import math
3  print(math.sqrt(16))

```

9. (Unreachable Code – Return Inside Loop)

Task:

Analyze given code where a return inside a loop prevents full iteration. Use AI to fix it.

```

# Bug: Early return inside loop
def total(numbers):
    for n in numbers:
        return n
print(total([1,2,3]))

```

Expected Output: Corrected code accumulates sum and returns after loop.

```

7.4.9.py > ...
1  # Bug: Early return inside loop
2  def total(numbers):
3      total_sum = 0
4      for n in numbers:
5          total_sum += n
6      return total_sum
7  print(total([1, 2, 3]))

```

10. (Name Error – Undefined Variable)

Task:

Analyze given code where a variable is used before being defined. Let AI detect and fix the error.

```
# Bug: Using undefined variable
def calculate_area():
    return length * width
print(calculate_area())
```

Requirements:

- Run the code to observe the error.
- Ask AI to identify the missing variable definition.
- Fix the bug by defining length and width as parameters.
- Add 3 assert test cases for correctness.

Expected Output :

```
param.py > ...
1  def calculate_area(length, width):
2      return length * width
3  print(calculate_area(5,3))
```

11. (Type Error – Mixing Data Types Incorrectly)

Task:

Analyze given code where integers and strings are added incorrectly. Let AI detect and fix the error.

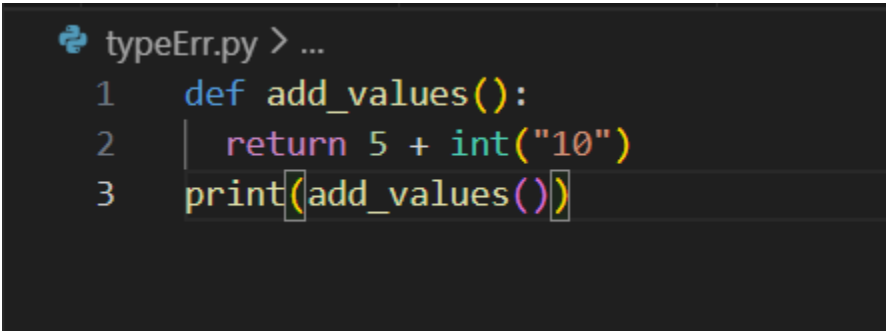
```
# Bug: Adding integer and string
def add_values():
    return 5 + "10"
print(add_values())
```

Requirements:

- Run the code to observe the error.
- AI should explain why int + str is invalid.
- Fix the code by type conversion (e.g., int("10") or str(5)).
- Verify with 3 assert cases.

Expected Output #6:

- Corrected code with type handling.
- AI explanation of the fix.
- Successful test validation.



```
typeErr.py > ...
1  def add_values():
2      return 5 + int("10")
3  print(add_values())
```

Explanation:

TypeError because of the different data types one should be converted

12. (Type Error – String + List Concatenation)

Task:

Analyze code where a string is incorrectly added to

```
# Bug: Adding string and list
def combine():
    return "Numbers: " + [1, 2, 3]
print(combine())
```

Requirements:

- Run the code to observe the error.
- Explain why str + list is invalid.
- Fix using conversion (str([1,2,3]) or " ".join()).
- Verify with 3 assert cases.

Expected Output:

- Corrected code
- Explanation
- Successful test validation

```
typeErr2.py > ...
1  def combine():
2      return "Numbers: " + str([1, 2, 3])
3  print(combine())
```

Explanation:

String and list cannot be concatenated together it should be explicitly changed

13. (Type Error – Multiplying String by Float)

Task: Detect and fix code where a string is multiplied by a float.

```
# Bug: Multiplying string by float
def repeat_text():
    return "Hello" * 2.5
print(repeat_text())
```

Requirements:

- Observe the error.
- Explain why float multiplication is invalid for strings.
- Fix by converting float to int.
- Add 3 assert test cases

```
C:\Users\lalit\OneDrive\Desktop\aiac\loopErr.py
1  def repeat_text():
2      return "Hello" * 2
3  print(repeat_text())
```

Explanation:

String repetition requires an integer count. Always convert floats explicitly using int() or round() based on your intent—never rely on implicit conversion.

Task 14 (Type Error – Adding None to Integer)

Task: Analyze code where None is added to an integer.

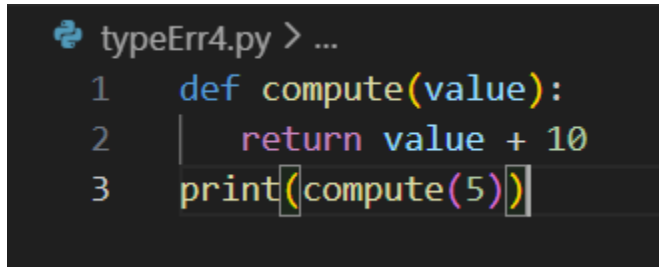
Bug: Adding None and integer

```
def compute():
```

```
value = None
return value + 10
print(compute())
```

Requirements:

- Run and identify the error.
- Explain why NoneType cannot be added.
- Fix by assigning a default value.
- Validate using asserts.



```
typeErr4.py > ...
1  def compute(value):
2      return value + 10
3  print(compute(5))
```

Task 15 (Type Error – Input Treated as String Instead of Number)

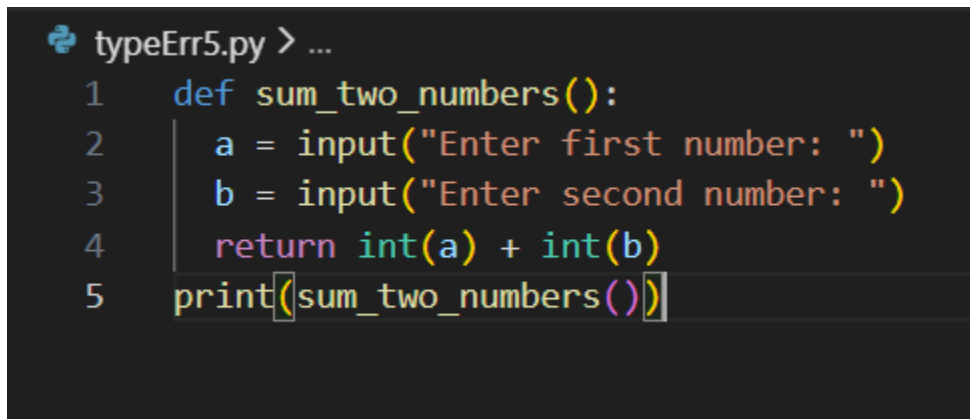
Task: Fix code where user input is not converted properly.

Bug: Input remains string

```
def sum_two_numbers():
a = input("Enter first number: ")
b = input("Enter second number: ")
return a + b
print(sum_two_numbers())
```

Requirements:

- Explain why input is always string.
- Fix using int() conversion.
- Verify with assert test cases.



```
typeErr5.py > ...
1  def sum_two_numbers():
2      a = input("Enter first number: ")
3      b = input("Enter second number: ")
4      return int(a) + int(b)
5  print(sum_two_numbers())
```